

next round: recalibrate $\tilde{C}$ from the merged native checkpoint

GENERATION

Native θ_t

Reference calibration

$$C_l = \frac{1}{M_l} \sum g_{iu} g_{iu}^\top$$

Soft spectral gain

$$T_{l,\tau} = [\mathbf{I} + \tau(\mathbf{I} - \tilde{C}_l)]^{-1}$$

$\tau = 1$; no direction zeroed

Fold in

$$W' = T^\top W$$
$$b' = T^\top b$$

Generate

1 raw completion
per prompt

raw completions — wrong and empty ones kept

TRAINING

Ordinary LoRA

all nonpadding targets
initialised from saved θ_t

Merge

native θ_{t+1}

Native evaluation

no spectral transform

saved
native θ_t

Solid = generation and training path · dashed = evaluation branch · dotted = saved-state restoration